Q1

1. (a) To estimate β_1, the researcher should use Fixed Effects estimator, which can prevent the influences of variables that not relative to time, like a_i, by demeaning the data:

$$\ddot{y}_{it} = y_{it} - \bar{y}_i$$

$$\ddot{x}_{it} = x_{it} - \bar{x}_i$$

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it} + \ddot{u}_{it}$$

then perform OLS regression to get β_1.

(b) soil quality: which not change through times from 2010 → 2019, differences in different villages will effect crop yield.

2. (a) γ_t represents time-specific unobserved effects that affect all the villages in a year from 2010-2019.

researcher should control time fixed effect, by including time dummies for each year in regression

$$Y_{it} = \beta_1 X_{it} + \sum_{t=2010}^{2019} \delta_t \text{Year}_t + u_{it}$$

this control the time-specific factors, remove its influence on X_{it}.

3. This method account for both time-specific effects and individual-specific effects by including γ_t and a_i

This can reduced the omitted variable bias and improved precision

4. suppose first demean by entity.

$$\ddot{Y}_{it} = Y_{it} - \bar{Y}_i \qquad \ddot{X}_{it} = X_{it} - \bar{X}_i$$

this removes fixed effect a_i

so we have $\ddot{Y}_{it} = \beta_1 \ddot{X}_{it} + \ddot{u}_{it}$

Then demeaning by time

$$\hat{\ddot{Y}}_{it} = \ddot{Y}_{it} - \bar{\ddot{Y}}_t \qquad \hat{\ddot{X}}_{it} = \ddot{X}_{it} - \bar{\ddot{X}}_t$$

Alternatively, first demean time than entity, the result is the same, because two operations are linear and commute, they are independent of each other, so whether we choose one of them first to demean, result is the same.

Q2

1- using DID

$$DiD = \left[\left(\frac{1}{n_1} \sum_{i:D_i=1} (Y_{i1} - Y_{i0}) \right) - \left(\frac{1}{n_0} \sum_{i:D_i=0} (Y_{i1} - Y_{i0}) \right) \right]$$

can be simplified to

$$DiD = (\bar{Y}_{1,1} - \bar{Y}_{1,0}) - (\bar{Y}_{0,1} - \bar{Y}_{0,0})$$

$\bar{Y}_{1,1}$ is average consumption after policy

$\bar{Y}_{1,0}$ is average consumption before policy

$\bar{Y}_{0,1}$ is average consumption in other provinces after policy

$\bar{Y}_{0,0}$ is average consumption in other provinces before policy.

2.

key assumption is that for DiD estimator to be an unbiased estimator of the treatment effect is the parallel trends assumption.

which $E[Y_{i1} - Y_{i0} | D_i = 1] =$

$= [Y_{i1} - Y_{i0} | D_i = 0]$

3. It may overestimate because the increased effect in Gansu households may lead to a higher consumption growth compared to other provinces.

This change is not due to policy but to the behavioral change.

Q3 1.

First, Because two places may have pre-existing differences in labor markets, this may influence employment levels independently of the minimum wage policy

second,

some factors uncorrelated with minimum wage change such as hurricane attack may influence

employment in both NJ and PA

2.

$$Empl_{it} = \beta_0 + \beta_1 Treatment_t + \beta_2 Group_i + \beta_3 (Treatment_t \times Group_i) + \gamma X_{it} + \varepsilon_{it}$$

$Empl_{it}$: Employment in restaurant i (NJ, PA) at time t

$Treatment_t$: Dummy variable (0 for before-policy 1 for after policy)

$Group_i$: Dummy variable (0 for PA, 1 for NJ)

Treatment$_t$ x Group i:

Interaction term indicating the

the treatment effect.

X_{it} : Control vector

3. It is the parallel trends

assumptions, which is

that before policy,

the employment trends in NJ

and PA would be the same

over time ..

4. result is

$\beta_1 = -2.823965$

$\beta_2 = -3.599658$

$\beta_3 = 3.561757$

$\therefore DiD = 3.566757$

Minimum wage increasing from $4.25 to $5.05 increases 3.566757 units of full time employees in NJ fast food restaurants compared with fast food restaurants in PA.

5.

Because DID method eliminates the potential interference of other factors that may affect employment by leveraging time variation before and after the policy and comparing differences between the treatment group and control group, so that under this situation, employment trends in NJ and PA are the same before policy.